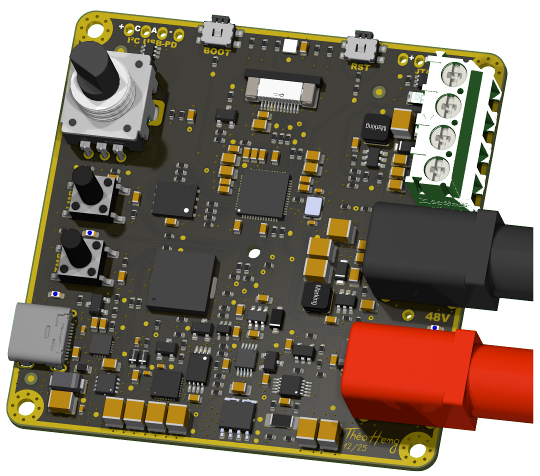
PD240W

Variant: RELEASED

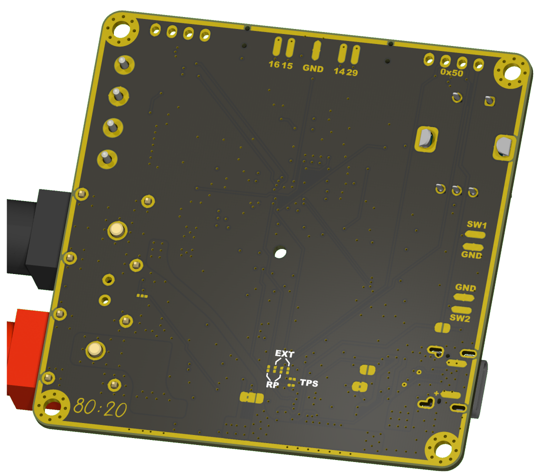
2026-03-15

Rev 1.0.0

TOP VIEW



BOTTOM VIEW



NOTES

An adjustable power supply using USB-C Power Delivery negotiation, supporting up to 240W at 48V 5A. This device is designed to be compatible with USB-PD 3.1 and above.

Not fitted components are marked as X

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Date: 15-Mar-2026

DESIGN CONSIDERATIONS

DESIGN NOTE: Example text for informational design notes.

DESIGN NOTE: Example text for debug notes.

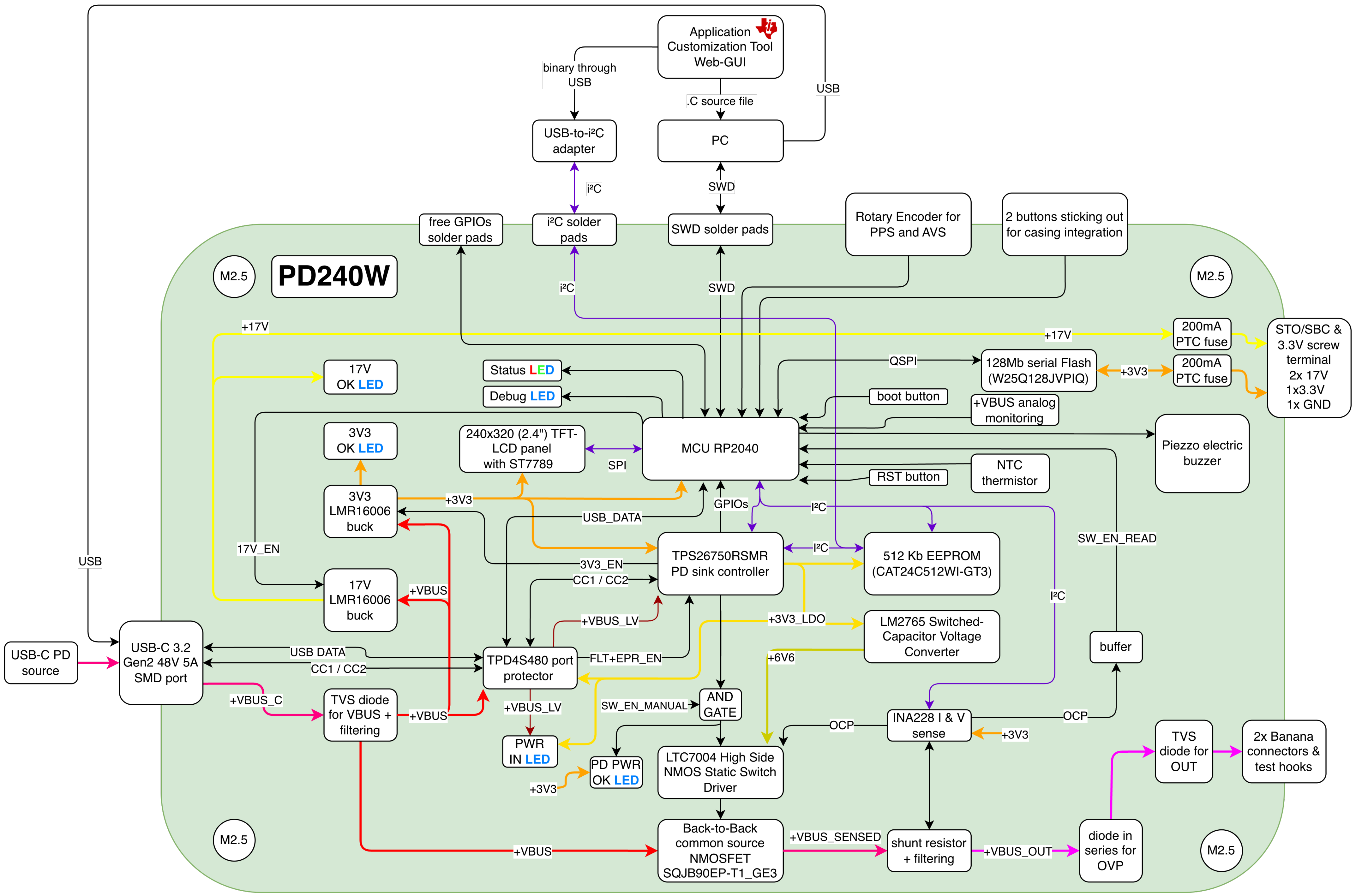
DESIGN NOTE: Example text for cautionary design notes.

DESIGN NOTE: Example text for critical design notes.

LAYOUT NOTE: Example text for critical layout guidelines.

Comments:	Company: Company Name		Variant: RELEASED	Git Hash: ae7b12e
	Board Name: PD240W		Project Name: PD240W	
Sheet Title:	File Name: PD240W_kicad.kicad_sch	Designer: Theo Heng	Date: 2026-02-27	Revision: 1.0.0
Sheet Path: /		Reviewer:	Size: A3	Sheet: 1 of 12

[2] Block Diagram



Target specifications:

Input voltage:	5 - 53 V
Output Voltage:	3.3 - 48 V
Max. Output Current:	5 A
Max Power Output:	240 W

Comments:	Company: Company Name		Variant: RELEASED	Git Hash: ae7b12e
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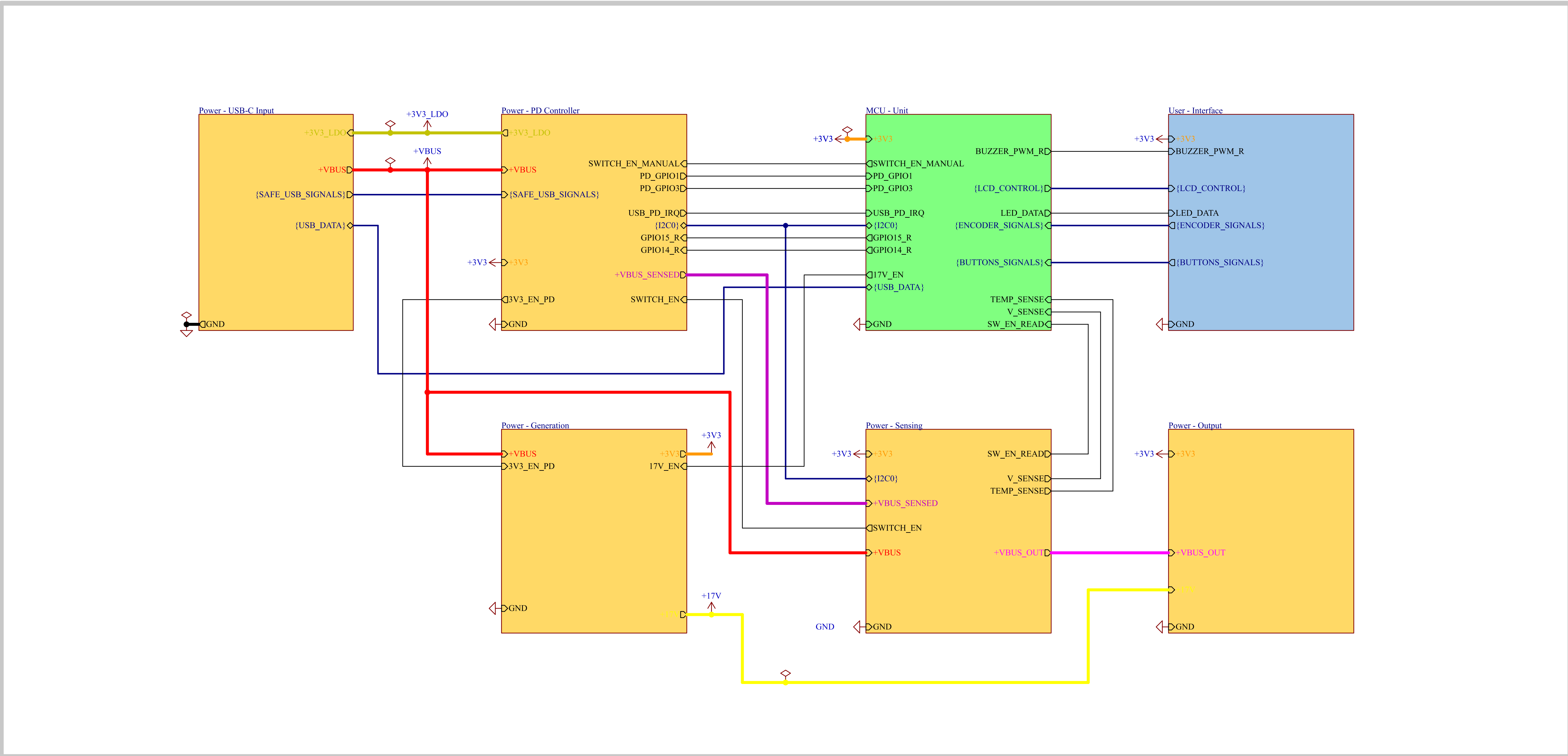
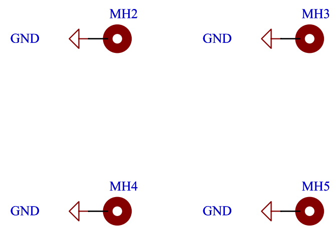
Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Theo Heng	Date: 2026-02-27	Revision: 1.0.0
Sheet Path: /Block Diagram/		Reviewer:	Size: A3	Sheet: 2 of 12

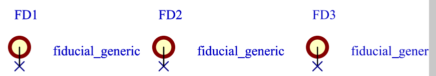
[3] Project Architecture





Mounting Holes M2.5

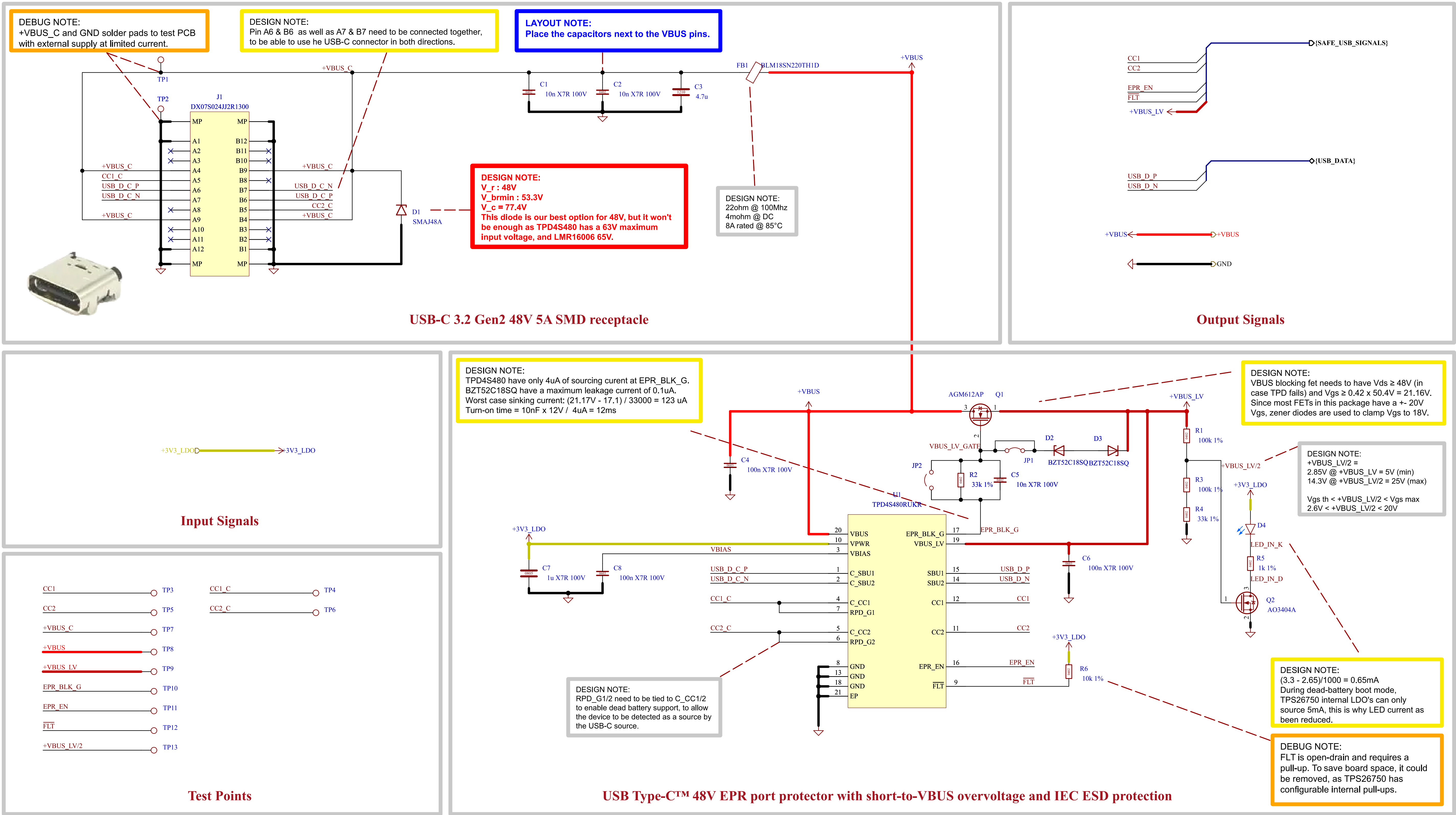
Casing Centering Hole



Fiducials - Top

Comments:	Company: Company Name		Variant: RELEASED	Git Hash: ae7b12e
	Board Name: PD240W		Project Name: PD240W	
Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Theo Heng	Date: 2026-02-27	Revision: 1.0.0
Sheet Path: /Project Architecture/		Reviewer:	Size: A3	Sheet: 3 of 12

[4] Power - USB-C Input



Comments:	Company:		Variant: RELEASED	Git Hash: ae7b12e
	Board Name: PD240W		Project Name: PD240W	
Sheet Title: Power - USB-C Input	File Name: power_usb_c_input.kicad_sch	Designer: Theo Heng	Date: 2026-02-27	Revision: 1.0.0
Sheet Path: /Project Architecture/Power - USB-C Input/		Reviewer:	Size: A3	Sheet: 4 of 12

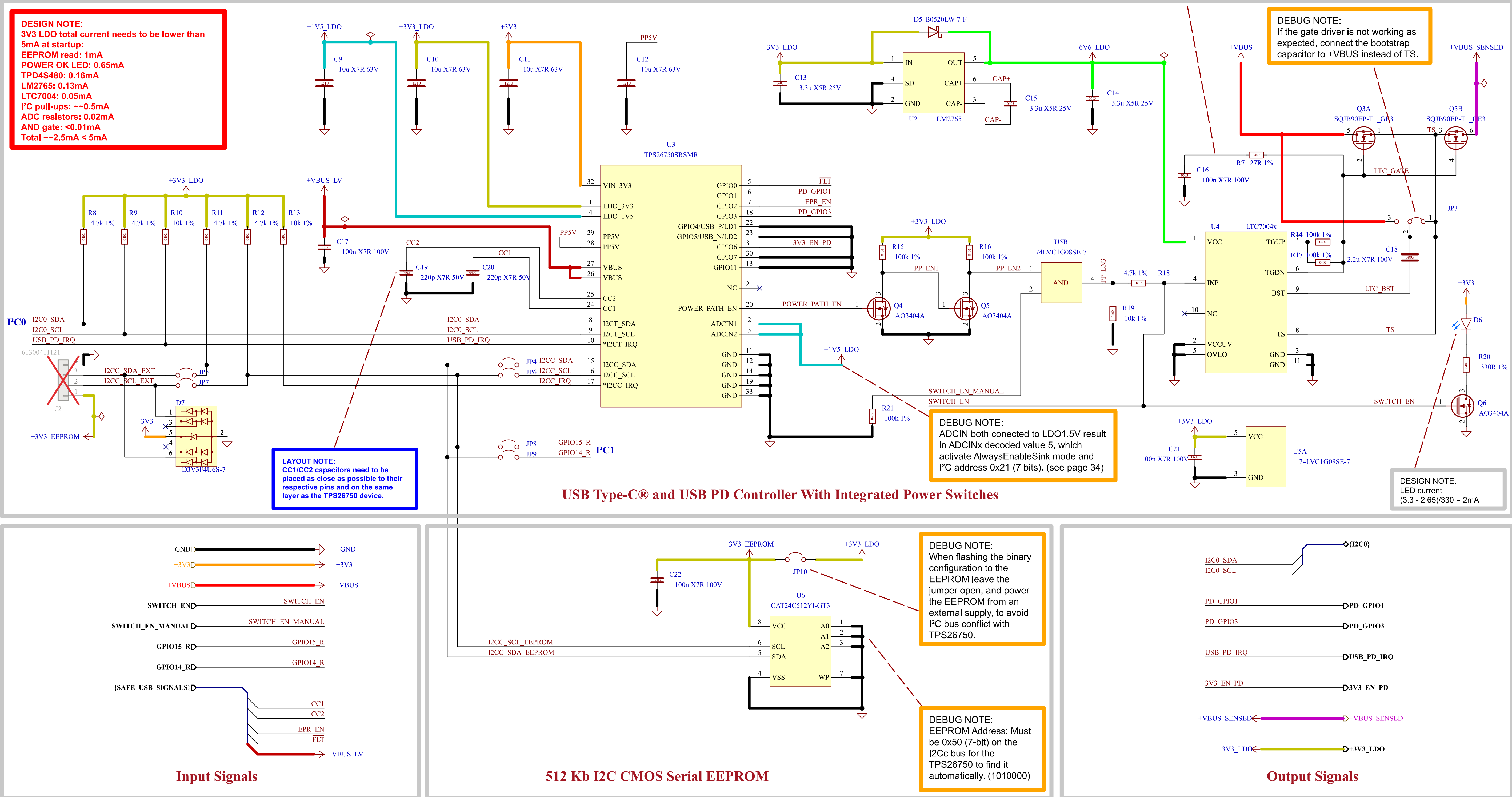
[5] Power - PD Controller

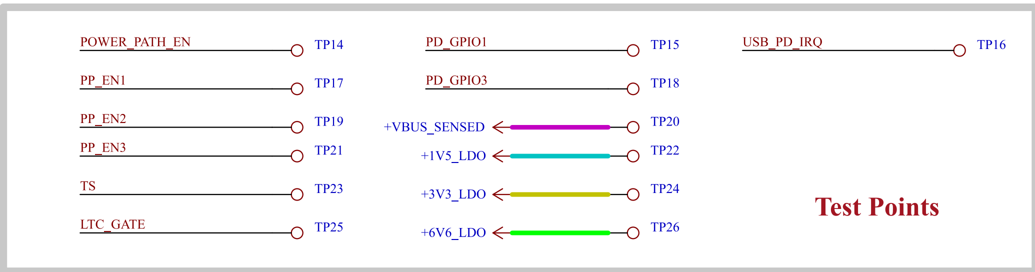
DESIGN NOTE:
The default state for a USB-C port (before or during negotiation, or if negotiation fails) is effectively USB 2.0 specs: 5V @ 500mA. Therefore we want to limit the inrush current. Synapticon drives have an averager of 150uF of input capacitance. As explained in application informations of LTC7004, the gate capacitor and TGUP resistors are chosen to limit the current to a safe limit of arround 250mA.

I_inrush = (0.7x12Vx150uF)/(50kOhm x 100nF) = 252mA

Time to reach 48V = (150uF x 48) / 0.252A = 29ms

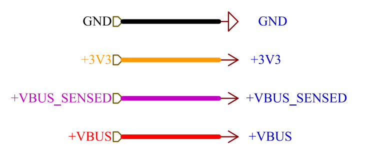
Boost capacitor adjustments: SQJB90EP has a max gate charge of 2x25nC = 50nC
The relationship for CB that needs to be maintained when CG is used is given by: Cb > (mosfet Qg)/1V + 10 x CG = 1.05 uF
With derating, a 2.2uF cap is necessary.





Comments:	Company:		Variant: RELEASED	Git Hash: ae7b12e
	Board Name: PD240W		Project Name: PD240W	
Sheet Title: Power - PD Controller	File Name: power_pd_controller.kicad_sch	Designer: Theo Heng	Date: 2026-02-27	Revision: 1.0.0
Sheet Path: /Project Architecture/Power - PD Controller/		Reviewer:	Size: A3	Sheet: 5 of 12

[6] Power - Sensing

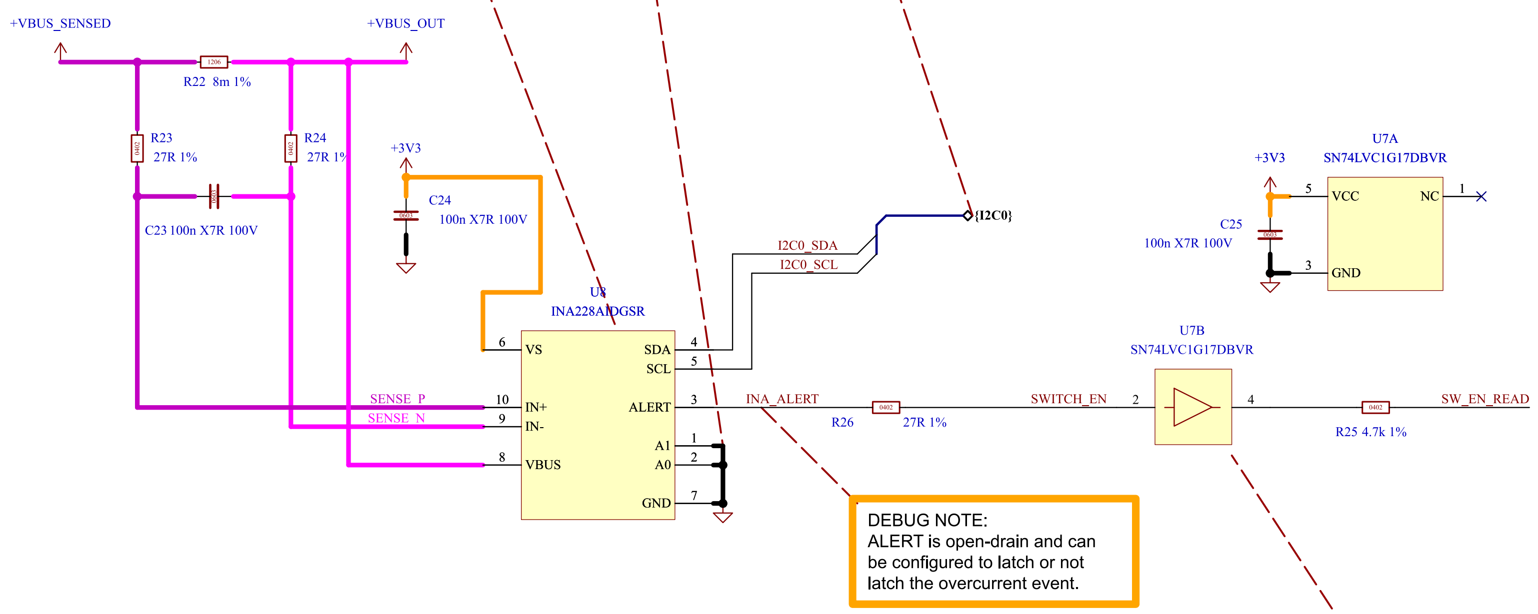


Input Signals

DESIGN NOTE:
ADCRANGE = 1 (+- 40.96 mV range)
V_{SENSE} = 5A x 0.008 = 40 mV. This utilizes 97.6% of the ADC range, providing the highest possible resolution.
Power Dissipation: P = 5² x 0.008 = 0.2W

DEBUG NOTE:
INA228AIDGSR address when A1 & A0 are tied to GND is 0x40 (7 bits 1000000) (see page 19)

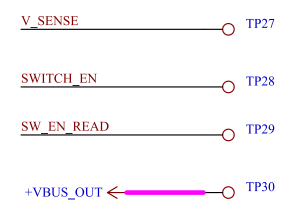
DESIGN NOTE:
I2C pullups are on Power - PD Conroller schematic



DEBUG NOTE:
ALERT is open-drain and can be configured to latch or not latch the overcurrent event.

DESIGN NOTE:
SN74LVC1G17 is a safety buffer to prevent software mistakes. Driving SWEN_READ cannot activate the power switch.

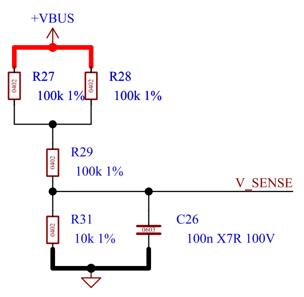
85-V, 20-Bit, Ultra-Precise Power/Energy/Charge Monitor With I²C Interface



Test Points

DESIGN NOTE:
Network ratio = 1/16 = 0.0625
@48V V_SENSE = 3V
@5V V_SENSE = 0.31V

LAYOUT NOTES:
Place capacitor close to MCU DC pin. Route away from noisy traces.

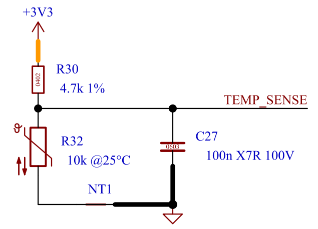


Input Voltage Sensing

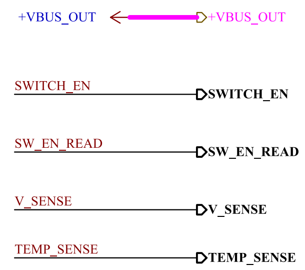
DESIGN NOTE:
B-Constant is 3950K
1.2k @ 80°C -> 0.67V
2.5k @ 60°C -> 1.14V
10k @ 25°C -> 2.24V
13k @ 20°C -> 2.42V

LAYOUT NOTES:
NTC must be placed close and thermally coupled to the ground plane near the relevant heat source. Route away from noisy traces.

DEBUG NOTE:
Low side measurement is used to improve thermal coupling, which invert the maths in software.



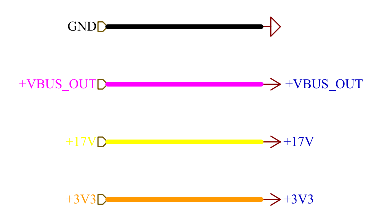
Temperature Sensing



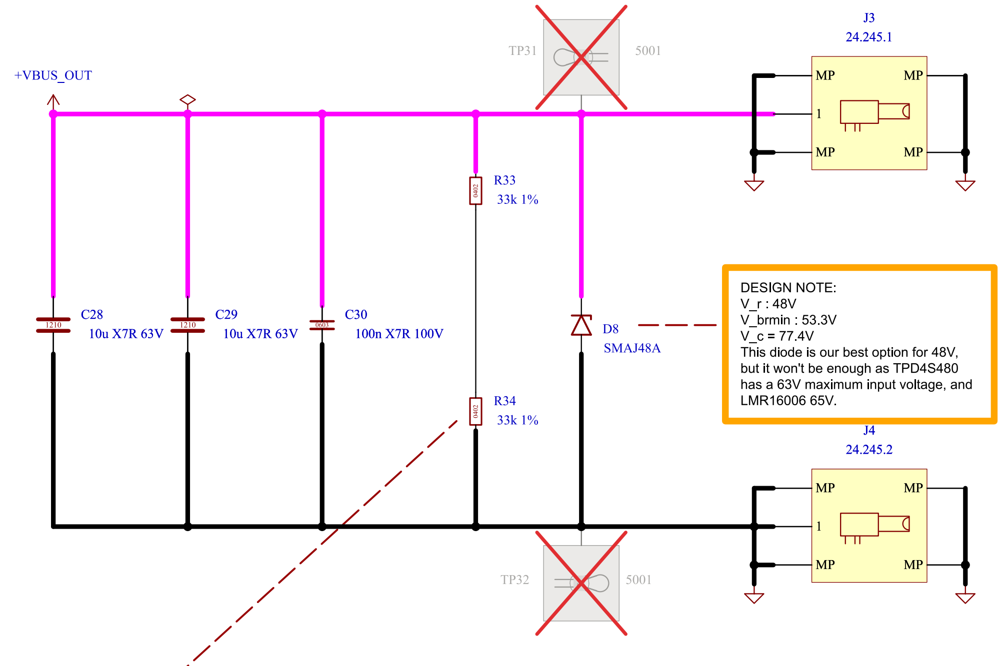
Output Signals

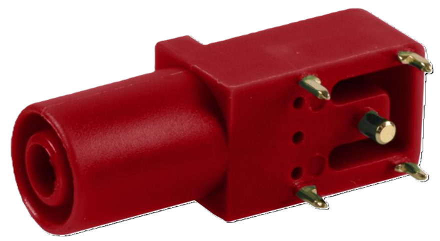
Comments:	Company:		Variant: RELEASED	Git Hash: ae7b12e
	Board Name: PD240W		Project Name: PD240W	
Sheet Title: Power - Sensing	File Name: power_sensing.kicad_sch	Designer: Theo Heng	Date: 2026-02-27	Revision: 1.0.0
Sheet Path: /Project Architecture/Power - Sensing/		Reviewer:	Size: A3	Sheet: 6 of 12

[7] Power - Output



Input Signals





FCR7350R - C7437326

LAYOUT NOTES:
Space Banana connectors by a standard 19.05mm.
Add mechanical stitching vias to +VBUS_OUT and GND to provide mechanical anchors against external torque applied to the connectors.

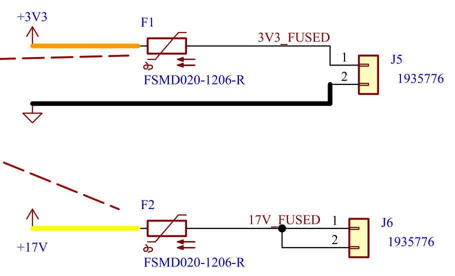
DESIGN NOTE:
V_r : 48V
V_brmin : 53.3V
V_c = 77.4V
This diode is our best option for 48V, but it won't be enough as TPD4S480 has a 63V maximum input voltage, and LMR16006 65V.

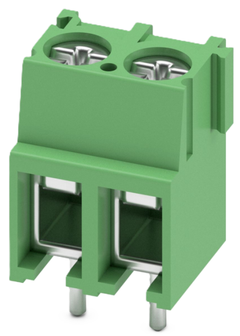
FCR7350B - C7437327

DESIGN NOTE:
RC = 66k x 20uF = 1.3ms
Safe to touch in 3 seconds
max power losses: 0.73mA

4mm Banana SMT RED & BLACK Output Connectors

DESIGN NOTE:
I_hold = 200mA
I_trip = 400mA

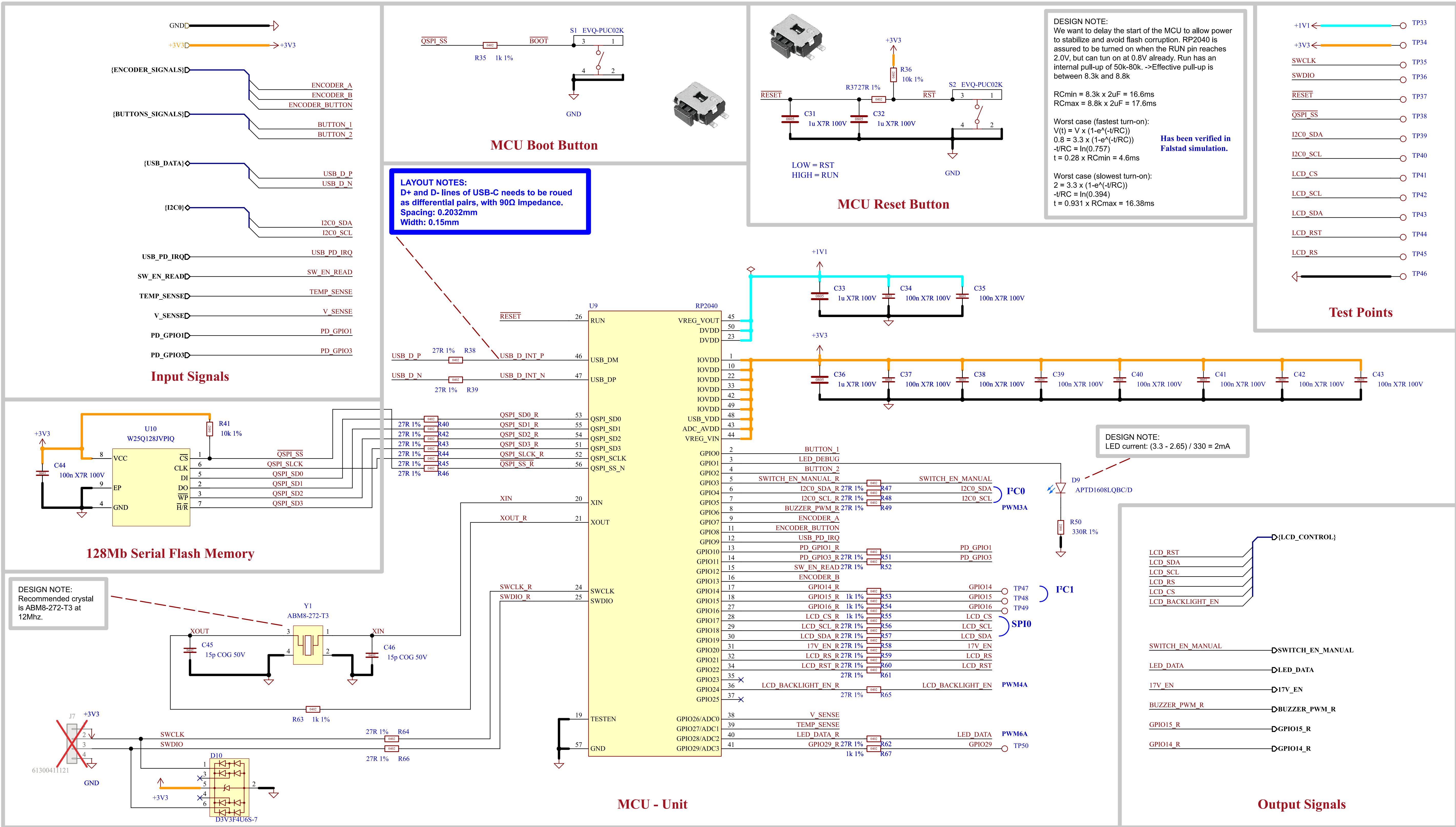




Screw Terminals for 17V

Comments:	Company:		Variant: RELEASED	Git Hash: ae7b12e
	Board Name: PD240W		Project Name: PD240W	
Sheet Title: Power - Output	File Name: power_output.kicad_sch	Designer: Theo Heng	Date: 2026-02-27	Revision: 1.0.0
Sheet Path: /Project Architecture/Power - Output/		Reviewer:	Size: A3	Sheet: 7 of 12

[8] MCU - Unit



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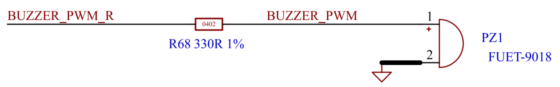
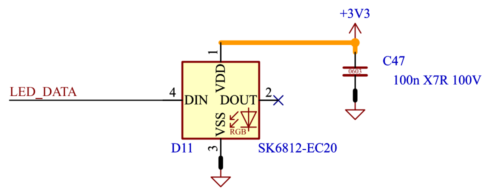
	Board Name: PD240W		Project Name: PD240W	
Sheet Title: MCU - Unit	File Name: mcu_unit.kicad_sch	Designer: Theo Heng	Date: 2026-02-27	Revision: 1.0.0
Sheet Path: /Project Architecture/MCU - Unit/		Reviewer:	Size: A3	Sheet: 8 of 12

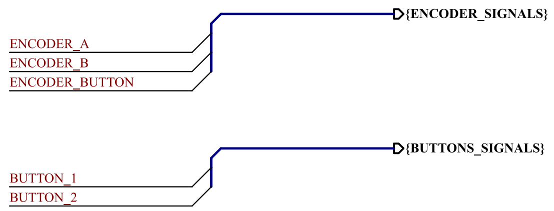
[9] User - Interface



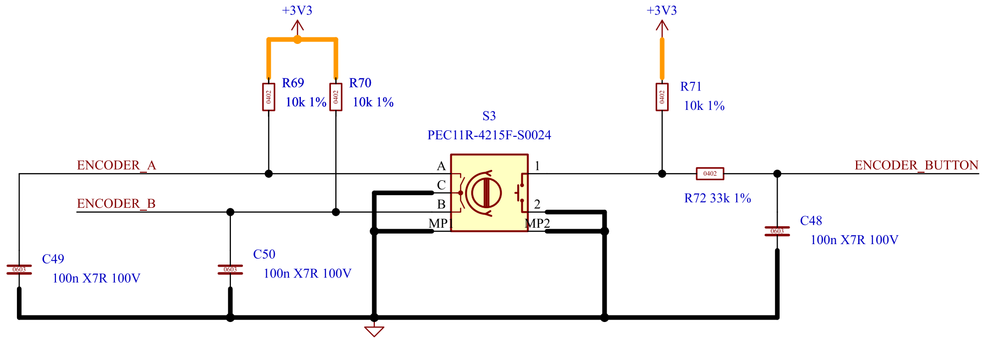
9x9mm SMD 65dB piezoelectric buzzer



Addressable RGD LED



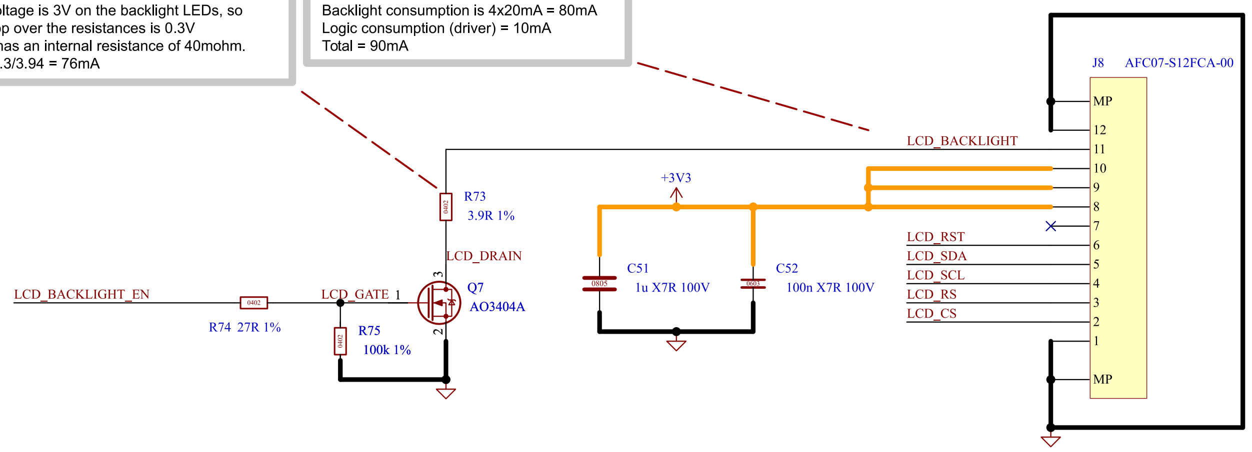
Output Signals



24 PPR Rotary Encoder with 24 Detents & Button

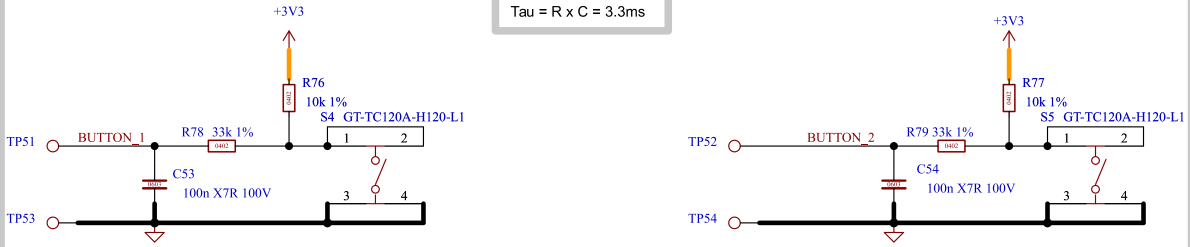
DESIGN NOTE:
Forward voltage is 3V on the backlight LEDs, so voltage drop over the resistances is 0.3V
AO3404A has an internal resistance of 40mohm.
current = 0.3/3.94 = 76mA

DESIGN NOTE:
Backlight consumption is 4x20mA = 80mA
Logic consumption (driver) = 10mA
Total = 90mA



12-pins, Low-side Contact FPC Connector and Backlight Driver

DESIGN NOTE:
Tau = R x C = 3.3ms

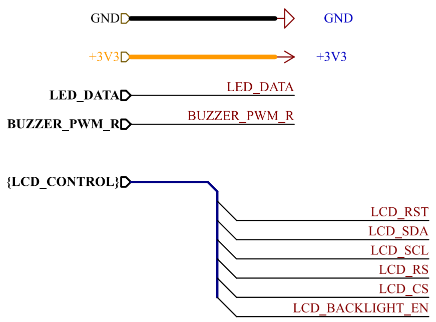


2x User Tactile Switches

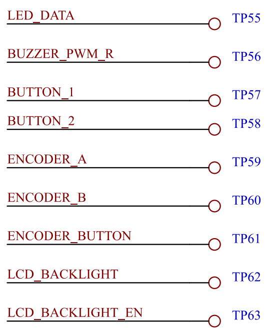
LAYOUT NOTES:
FPC connector of LCD display is 20.7mm long, which allows for arround 10mm of clearance between the top of the PCB and the bottom of the display.

NO.	SYMBOL
1	GND
2	CS
3	RS
4	SCL
5	SDA
6	RESET
7	NC
8	IOVCC
9	VCC
10	LEDA
11	LEDK
12	GND

240x320 (2.4") TFT-LCD Panel with ST7789 driver & FPC

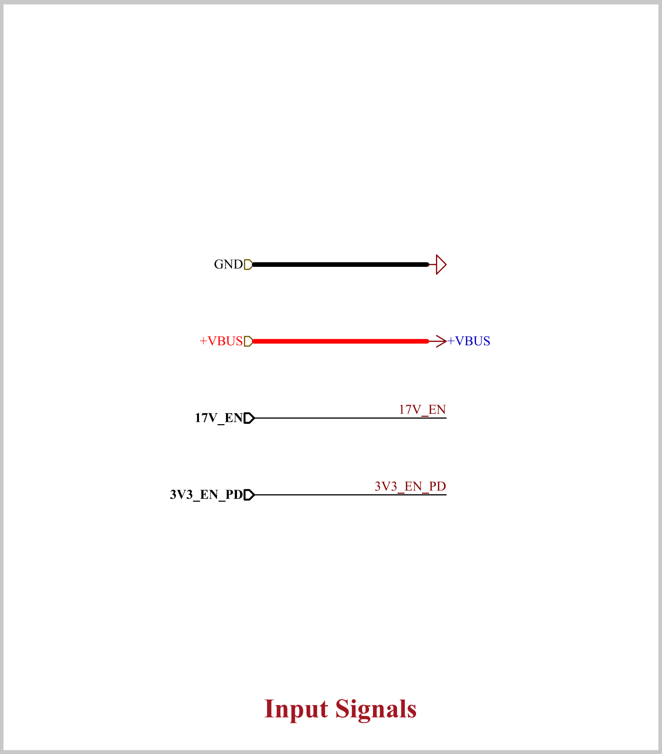


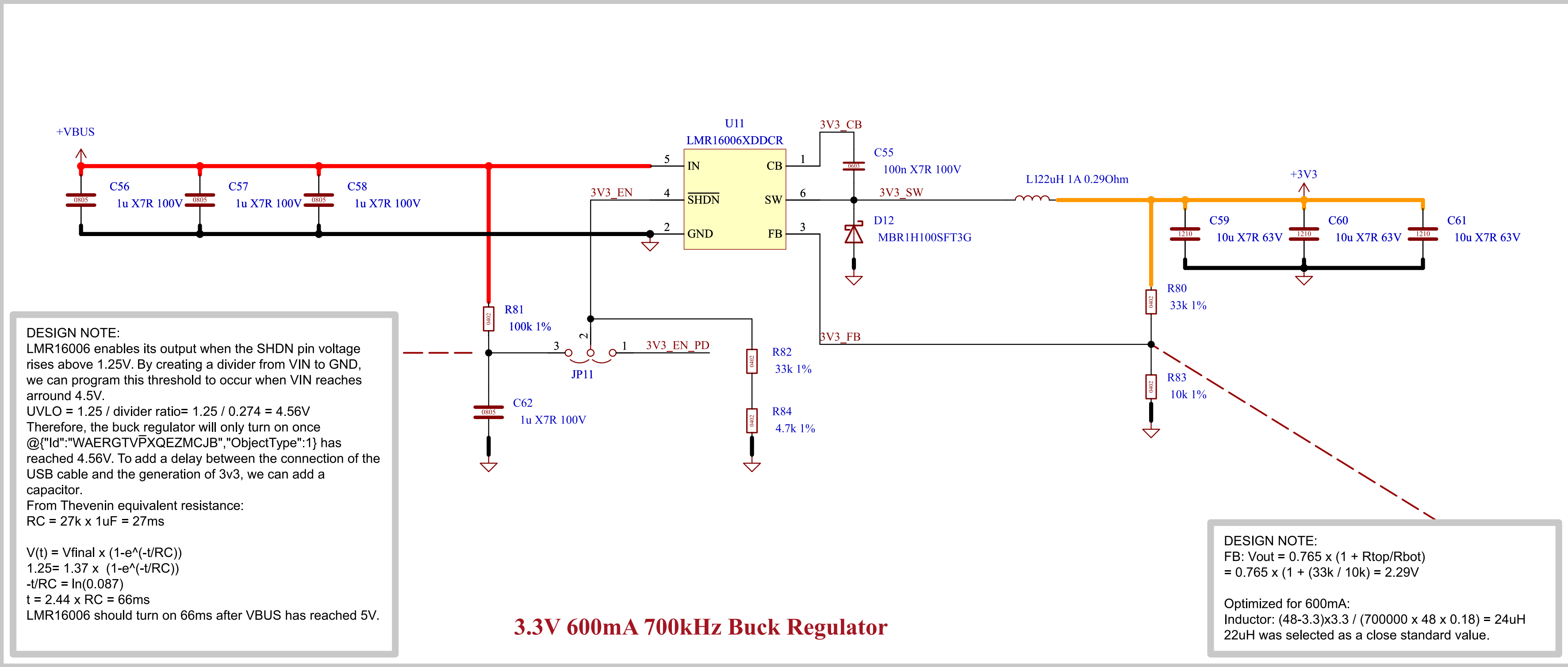
Input Signals



Test Points

Comments:	Company:		Variant:	Git Hash:
			RELEASED	ae7b12e
	Board Name: PD240W		Project Name: PD240W	
Sheet Title: User - Interface	File Name: user_interface.kicad_sch	Designer: Theo Heng	Date: 2026-02-27	Revision: 1.0.0
Sheet Path: /Project Architecture/User - Interface/		Reviewer:	Size: A3	Sheet: 9 of 12



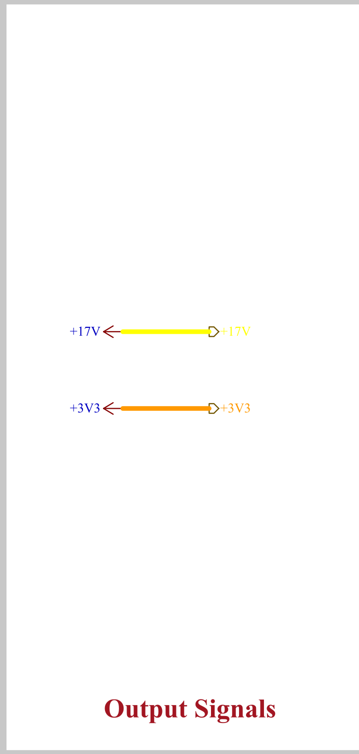


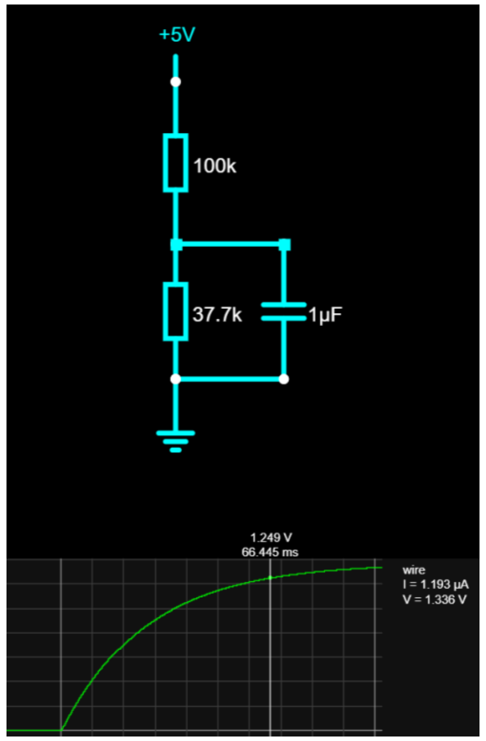
DESIGN NOTE:
LMR16006 enables its output when the SHDN pin voltage rises above 1.25V. By creating a divider from VIN to GND, we can program this threshold to occur when VIN reaches arround 4.5V.
UVLO = 1.25 / divider ratio= 1.25 / 0.274 = 4.56V
Therefore, the buck regulator will only turn on once @{"Id":"WAERGTVPXQEZMCJB","ObjectType":1} has reached 4.56V. To add a delay between the connection of the USB cable and the generation of 3v3, we can add a capacitor.
From Thevenin equivalent resistance:
RC = 27k x 1uF = 27ms

V(t) = Vfinal x (1-e^(-t/RC))
1.25= 1.37 x (1-e^(-t/RC))
-t/RC = ln(0.087)
t = 2.44 x RC = 66ms
LMR16006 should turn on 66ms after VBUS has reached 5V.

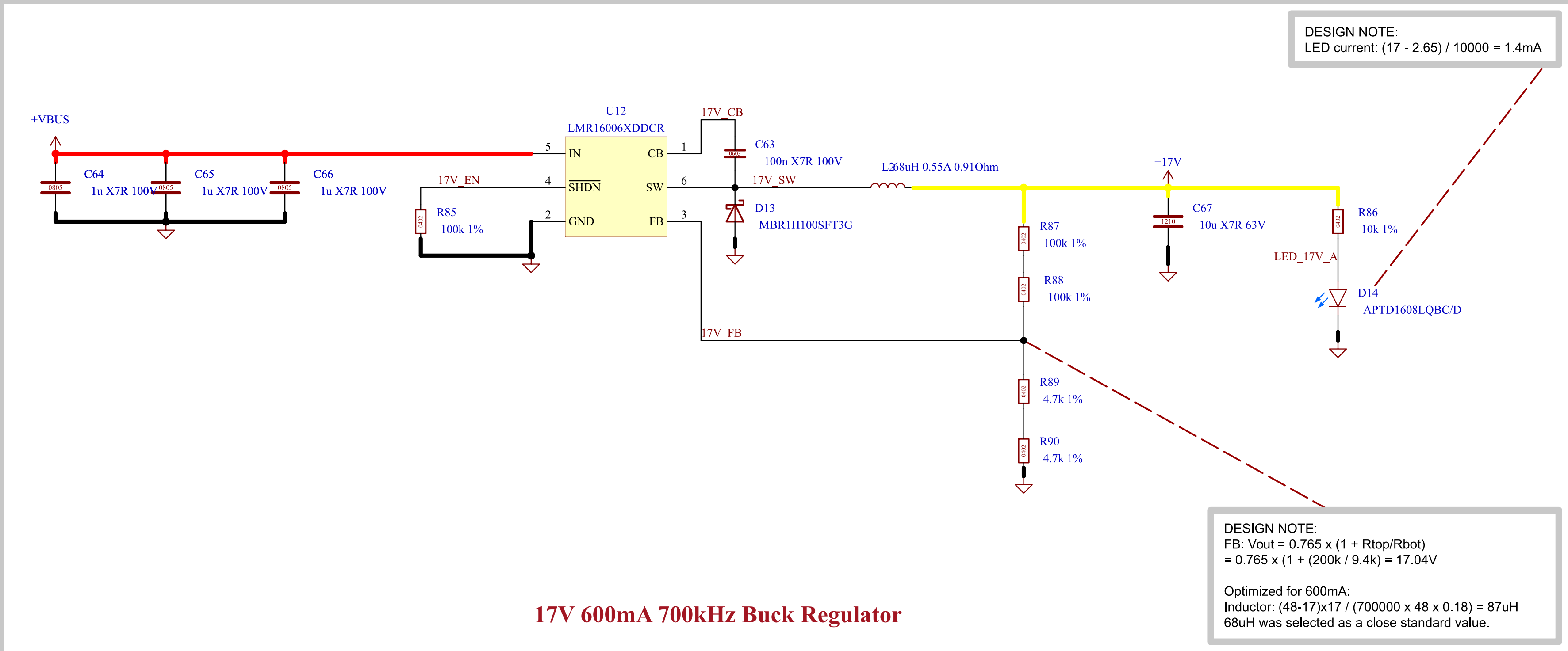
DESIGN NOTE:
FB: Vout = 0.765 x (1 + Rtop/Rbot)
= 0.765 x (1 + (33k / 10k) = 2.29V

Optimized for 600mA:
Inductor: (48-3.3)x3.3 / (700000 x 48 x 0.18) = 24uH
22uH was selected as a close standard value.





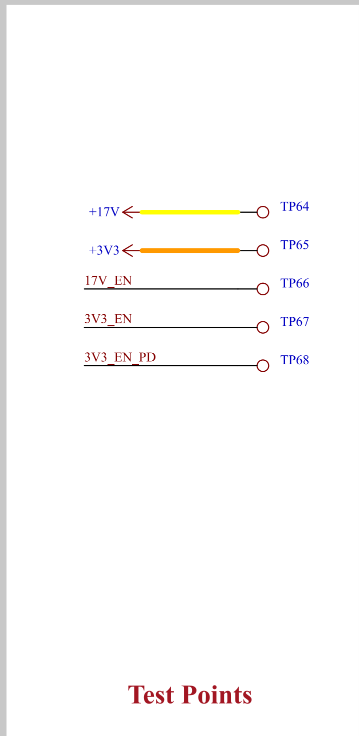
Falstad Simulation Results: 1.25V reached in 66ms.



DESIGN NOTE:
LED current: (17 - 2.65) / 10000 = 1.4mA

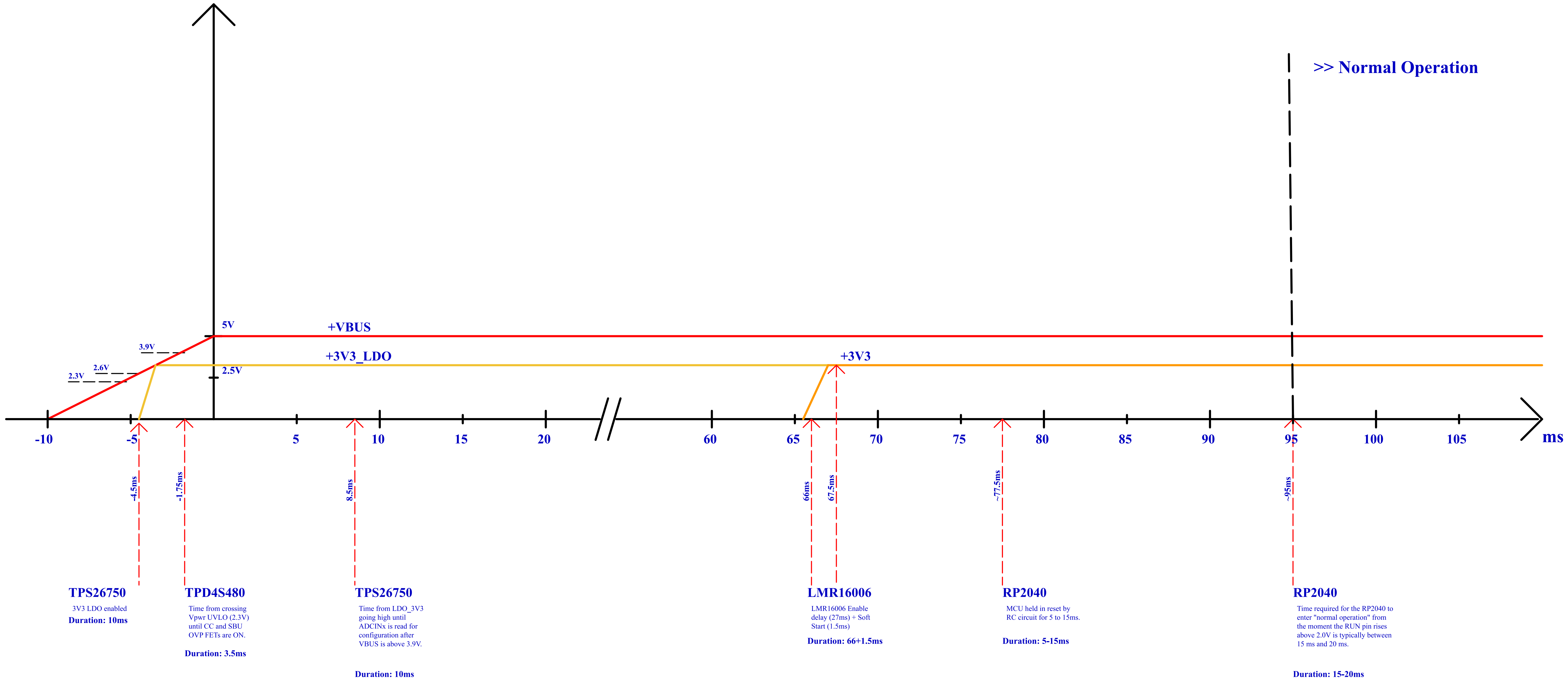
DESIGN NOTE:
FB: Vout = 0.765 x (1 + Rtop/Rbot)
= 0.765 x (1 + (200k / 9.4k) = 17.04V

Optimized for 600mA:
Inductor: (48-17)x17 / (700000 x 48 x 0.18) = 87uH
68uH was selected as a close standard value.



Comments:	Company:		Variant: RELEASED	Git Hash: ae7b12e
	Board Name: PD240W		Project Name: PD240W	
Sheet Title: Power - Generation	File Name: power_generation.kicad_sch	Designer: Theo Heng	Date: 2026-02-27	Revision: 1.0.0
Sheet Path: /Project Architecture/Power - Generation/		Reviewer:	Size: A3	Sheet: 10 of 12

[11] Power - Sequencing



Comments:	Company: Company Name		Variant: RELEASED	Git Hash: ae7b12e
	Board Name: PD240W		Project Name: PD240W	
Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: Theo Heng	Date: 2026-02-27	Revision: 1.0.0
Sheet Path: /Power - Sequencing/		Reviewer:	Size: A3	Sheet: 11 of 12

[12] Revision History

Comments:	Company: Company Name		Variant: RELEASED	Git Hash: ae7b12e
	Board Name: PD240W		Project Name: PD240W	
Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Theo Heng	Date: 2026-02-27	Revision: 1.0.0
Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 12 of 12